1. Create a Sales Data Warehouse

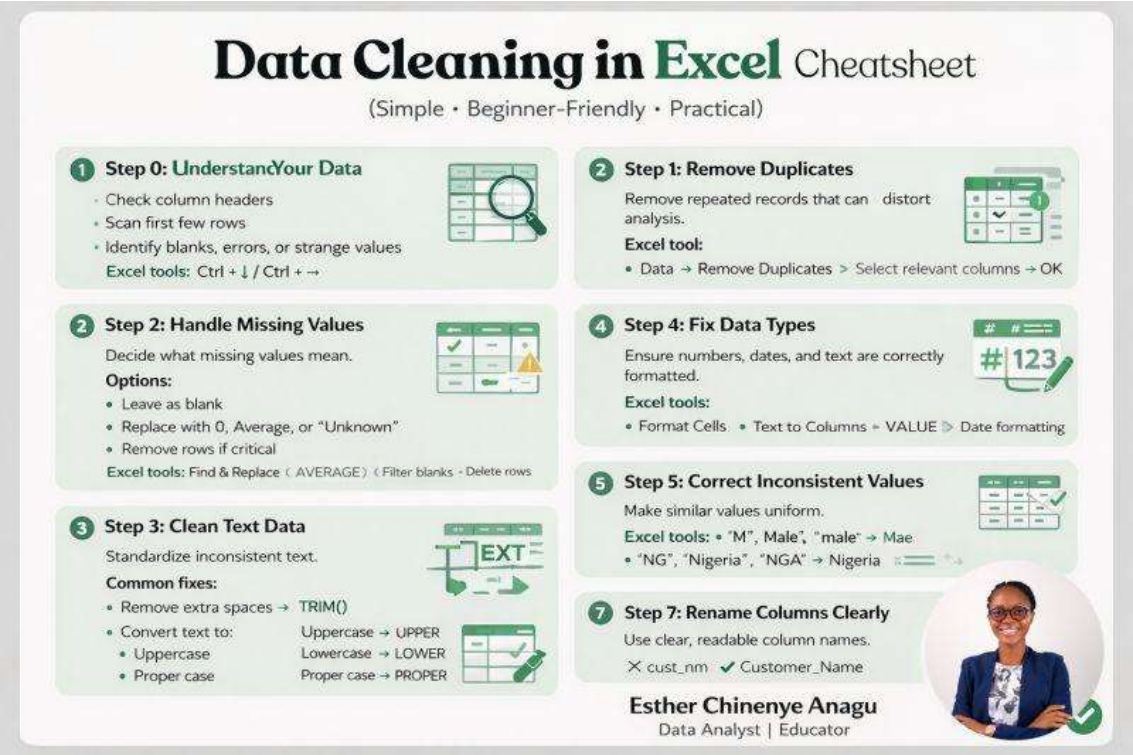
Unit I – Data Warehousing

Task: Create a small sales dataset with 10 records in Excel.

The student should:

- Create columns for Product, City, Month, Quantity and Sales Amount.
- Identify the fact table and dimension tables.
- Draw a simple star schema.
- Write two questions that the warehouse can answer.

Time: 30 minutes



2. Data Cleaning Challenge

Unit II – Data Preprocessing

Task: Prepare a table containing 10 customer records, including missing values, duplicate records and spelling errors.

The student should:

- Identify data quality problems.
- Remove duplicates.
- Handle missing values.
- Submit the corrected dataset.

Time: 20 minutes

Answers

1.)Create a Sales Data Warehouse:

1. Aim

To design and create a small sales data warehouse using a dataset containing 10 sales records in Microsoft Excel, identify the fact table and dimension tables, construct a star schema, and formulate analytical questions that can be answered using the data warehouse.

2. Introduction

A data warehouse is a centralized repository that collects, integrates, and stores structured data from different sources for reporting, analysis, and decision-making. It helps organizations understand their business performance by organizing historical information in a meaningful format.

In a sales organization, large amounts of data are generated through product sales, customer purchases, branch transactions, and monthly revenue. Analyzing these records individually can be difficult. A sales data warehouse organizes this information so that managers can identify popular products, compare city-wise sales, monitor monthly revenue, and make better business decisions.

A sales data warehouse commonly uses a fact table and several dimension tables. The fact table contains numerical measurements, such as quantity sold and sales amount. Dimension tables provide descriptive information about products, cities, and time periods.

A star schema connects the central fact table directly to the dimension tables. This structure makes sales information easier to understand, query, and analyze.

3. Objectives

The main objectives of this activity are:

1. To create a sales dataset containing 10 records using Microsoft Excel.
2. To organize sales information using Product, City, Month, Quantity, and Sales Amount columns.
3. To identify the fact table and its measurable attributes.

4. To create product, city, and month dimension tables.
5. To design a star schema showing the relationships between the tables.
6. To formulate business questions that can be answered using the warehouse.
7. To understand how data warehousing supports business intelligence and decision-making.

4. Tools Required

- Microsoft Excel for creating and organizing the dataset.
- A notebook or diagramming tool for drawing the star schema.
- Basic knowledge of tables, primary keys, foreign keys, and data analysis.

5. Procedure

Step 1: Create the sales dataset

Open Microsoft Excel and create the following column headings:

- Product
- City
- Month
- Quantity
- Sales Amount

Enter the following 10 sample sales records.

S. No.	Product	City	Month	Quantity	Sales Amount (₹)	Total Sales
1	Laptop	Chennai	January	5	₹2,50,000	
2	Mobile	Bangalore	January	10	₹1,50,000	
3	Tablet	Hyderabad	February	8	₹1,20,000	Total Quantity
4	Laptop	Mumbai	February	3	₹1,50,000	
5	Headphones	Chennai	March	15	₹30,000	
6	Mobile	Hyderabad	March	12	₹1,80,000	
7	Keyboard	Bangalore	April	20	₹20,000	
8	Laptop	Chennai	April	4	₹2,00,000	
9	Tablet	Mumbai	May	6	₹90,000	
10	Headphones	Bangalore	May	10	₹20,000	

The dataset contains different products, cities, months, quantities, and sales amounts. It can be used to demonstrate how a data warehouse organizes and analyzes business transactions.

Step 2: Understand the dataset

Each column represents a specific attribute of a sales transaction.

Column	Description	Purpose
Product	Name of the product sold	Identifies the item sold
City	City where the sale occurred	Supports geographical analysis
Month	Month in which the sale occurred	Supports time-based analysis
Quantity	Number of units sold	Measures sales volume
Sales Amount	Total monetary value of the sale	Measures revenue

For example, the first record indicates that five laptops were sold in Chennai during January, generating ₹250,000 in sales.

Step 3: Identify the fact table

The Sales_Fact table is the central table of the warehouse. It stores measurable information about sales transactions and keys that connect each transaction to its descriptive dimension records.

Its main measures are:

- Quantity: The number of units sold in a transaction.
- Sales Amount: The monetary value generated by the transaction.

In a relational implementation, the fact table also contains foreign keys that refer to the product, city, and month dimension tables.

Sales_Fact table

Product_ID	City_ID	Month_ID	Quantity	Sales Amount (₹)
P01	C01	M01	5	2,50,000
P02	C02	M01	10	1,50,000
P03	C03	M02	8	1,20,000
P01	C04	M02	3	1,50,000
P04	C01	M03	15	30,000
P02	C03	M03	12	1,80,000
P05	C02	M04	20	20,000
P01	C01	M04	4	2,00,000
P03	C04	M05	6	90,000
P04	C02	M05	10	20,000

Here, P01 identifies Laptop, C01 identifies Chennai, and M01 identifies January. These identifiers reduce repeated descriptive information and connect the fact table to the corresponding dimension records.

Step 4: Identify the dimension tables

Dimension tables provide descriptive details that help users understand and categorize the numerical information in the fact table.

A. Product_Dimension

This table stores product-related information. It allows managers to compare sales across different products.

Product_ID (Primary Key)	Product_Name
P01	Laptop
P02	Mobile
P03	Tablet
P04	Headphones
P05	Keyboard

B. City_Dimension

This table stores geographical information about where the sales transactions occurred.

City_ID (Primary Key)	City_Name
C01	Chennai
C02	Bangalore
C03	Hyderabad
C04	Mumbai

C. Month_Dimension

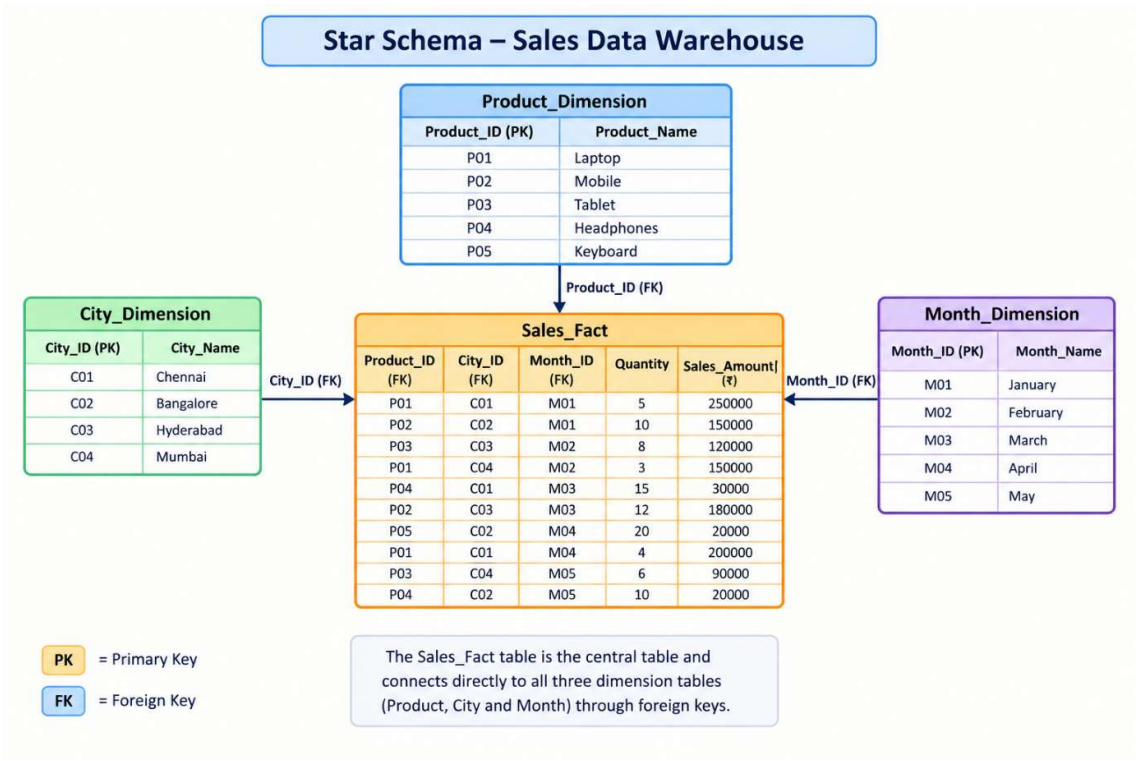
This table stores time-related information that helps analyze sales across different months.

Month_ID	Month_Name
M01	January
M02	February
M03	March
M04	April
M05	May

The product, city, and month identifiers are primary keys in their respective dimension tables. The same identifiers act as foreign keys in the Sales_Fact table.

Step 5: Draw the star schema

A star schema is a dimensional data model in which one central fact table connects directly to multiple dimension tables.



Conceptual representation of the four tables. In the actual schema, all three dimension tables connect directly to Sales_Fact.

Relationship explanation:

- Product_Dimension connects to Sales_Fact through Product_ID.
- City_Dimension connects to Sales_Fact through City_ID.
- Month_Dimension connects to Sales_Fact through Month_ID.

PK means primary key, and FK means foreign key. Each dimension record can be associated with multiple sales transactions.

Step 6: Calculate basic sales statistics

The dataset can be used to calculate total sales, total quantity, and product-wise revenue.

Total sales amount

$$\begin{aligned} \text{Total Sales} &= \sum \text{Sales Amount}\\ &= ₹1,210,000 \end{aligned}$$

Total quantity sold

$$\begin{aligned} \text{Total Quantity} &= \sum \text{Quantity}\\ &= 93\text{ units} \end{aligned}$$

The warehouse therefore contains ₹12.1 lakh in sales revenue across 93 units sold.

S. No.	Product	City	Month	Quantity	Sales Amount (₹)	Total Sales
1	Laptop	Chennai	January	5	₹2,50,000	
2	Mobile	Bangalore	January	10	₹1,50,000	
3	Tablet	Hyderabad	February	8	₹1,20,000	Total Quantity
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This analysis shows that Laptop generates the highest revenue, while Headphones has the lowest revenue among the listed products. Headphones nevertheless has a higher quantity sold than Laptop, demonstrating why revenue and sales volume are different measures.

Step 7: Formulate business questions

The sales warehouse can answer several questions that help managers understand performance.

Question 1: Which product generates the highest total sales amount?

The fact table can be grouped by Product using the Product_Dimension table. Sales Amount is then summed for each product.

Based on the sample data, Laptop generates the highest total sales amount of ₹600,000. This helps the company identify products that contribute most to revenue.

Question 2: Which city records the highest total sales amount during a particular month?

The fact table can be grouped by City and Month. The Sales Amount measure can then be summed for each city-month combination.

For example, Chennai records ₹250,000 in January and ₹200,000 in April. This type of analysis helps the company compare geographical performance and identify stronger sales periods.

Step 8: Advantages of the sales data warehouse

1. Better decision-making: Managers can use sales figures to plan business activities.
2. Historical analysis: Sales from different months can be compared.
3. Product performance analysis: Revenue and quantity can be measured for each product.
4. Geographical analysis: City-wise performance can be evaluated.
5. Simplified reporting: Structured tables make it easier to generate reports.
6. Improved data organization: Fact and dimension tables separate measurements from descriptive information.
7. Business intelligence support: The warehouse can later support charts, dashboards, and more advanced analytical queries.

Step 9: Applications

A sales data warehouse can be used in retail shops, supermarkets, e-commerce platforms, electronics stores, distribution companies, and other businesses that track product sales. It can support inventory planning, revenue forecasting, marketing analysis, and sales performance reporting.

Result

A sales data warehouse was designed using 10 sample sales records. The Sales_Fact table and the Product_Dimension, City_Dimension, and Month_Dimension tables were identified. A star schema was constructed, and business questions were formulated.

2.)Data Cleaning Challenge

1. Aim

To prepare a customer dataset containing 10 records with missing values, duplicate records, and spelling errors; identify data quality problems; apply appropriate data cleaning techniques; and submit a corrected dataset suitable for further data analysis.

2. Introduction

Data preprocessing is an important step in data analysis and data mining. Real-world datasets often contain incomplete, inaccurate, inconsistent, or repeated information. These problems can affect the accuracy of analytical results and lead to incorrect conclusions.

Data cleaning is the process of identifying and correcting these problems to improve the quality and reliability of data. It includes handling missing values, removing duplicate records, correcting spelling errors, standardizing formats, and validating the final dataset.

For example, if a customer database contains the city names “Bangalore,” “Banglore,” and “Bengaluru,” the records may be grouped incorrectly if the naming conventions are not standardized. Similarly, duplicate customer records may cause a business to overestimate the number of customers.

In this activity, a customer dataset containing 10 records is created. It includes missing values, a duplicate customer entry, and spelling errors. These issues are identified and corrected to produce a clean dataset.

3. Objectives

The main objectives of this activity are:

1. To prepare a sample customer dataset containing 10 records.
2. To identify missing values and incomplete records.
3. To detect duplicate customer entries.
4. To identify and correct spelling errors.
5. To apply appropriate techniques for handling missing values.
6. To remove duplicate records without unnecessarily losing valid information.
7. To produce a corrected dataset with consistent values.
8. To understand the importance of data quality in data preprocessing.

4. Tools Required

- Microsoft Excel for creating and cleaning the dataset.
- A computer or laptop.
- Basic knowledge of tables, cells, sorting, filtering, and duplicate detection.

5. Original Customer Dataset

Create a table in Microsoft Excel with the following columns: Customer ID, Customer Name, City, and Age.

Customer ID	Customer ID	City	Age
101	Arun	Chennai	22
102	Priya	Banglore	25
103	Kumar	Chennai	24
104	Divya	Hydrabad	23
105	Rahul	Mumbai	28
106	Sneha	Chennai	—
107	Vijay	Bangalore	26
108	Meena	—	27
109	Priya	Banglore	25
110	Karthik	Mumbai	30

This dataset contains four types of data quality issues: missing values, duplicate records, spelling errors, and inconsistent or incomplete information.

6. Identify Data Quality Problems

The first step in data cleaning is to examine the original dataset and identify the problems present in it.

6.1 Missing values

A missing value occurs when a field does not contain the expected information.

In the sample dataset:

- Customer 106, Sneha, has no age value.
- Customer 108, Meena, has no city value.

Effects of missing values:

- They can reduce the completeness of the dataset.
- They may prevent certain calculations or comparisons.
- They can affect statistical analysis.
- They may reduce the usefulness of reports if important fields are incomplete.

Solution: Replace a missing numerical value using an appropriate statistical method, such as the median, when justified. For missing categorical information, use a suitable category such as “Unknown” if the correct value cannot be established.

6.2 Duplicate records

Duplicate records occur when the same real-world information is recorded more than once.

In the sample dataset, customers 102 and 109 have the same name, city, and age:

Customer ID	Customer Name	City	Age
102	Priya	Banglore	25
109	Priya	Banglore	25

These records are treated as duplicates for this exercise because the customer details are identical. However, in a real database, matching names and ages alone do not always prove that two records belong to the same person. Additional identifiers or verification may be required.

Effects of duplicate records:

- They can inflate customer counts.
- They may distort statistical calculations.
- They can lead to duplicate communications or transactions.

- They can reduce the reliability of reports.

Solution: Identify duplicates using appropriate identifying fields and remove the redundant record after confirming that it represents the same customer.

6.3 Spelling errors

Spelling errors occur when a value is entered incorrectly.

The original dataset contains the following errors:

Incorrect Value	Correct Value
Banglore	Bangalore
Hydrabad	Hyderabad

These errors may cause the same city to appear as two separate categories during grouping or analysis.

Solution: Standardize spelling and naming conventions across the dataset.

6.4 Summary of data quality problems

Problem Type	Record	Problem	Action Required
Missing value	106	Age is blank	Replace with median age
Missing value	108	City is blank	Use “Unknown”
Duplicate record	109	Priya's details repeat record 102	Remove redundant record
Spelling error	102, 109	Banglore	Correct to Bangalore
Spelling error	104	Hydrabad	Correct to Hyderabad

7. Data Cleaning Procedure

Step 1: Inspect the dataset

Examine all rows and columns carefully. Look for empty cells, repeated records, misspelled words, inconsistent formats, and invalid values.

In Excel, sorting and filtering can help reveal missing or inconsistent entries.

Step 2: Detect duplicate records

Compare the customer details to identify repeated records. For this activity, Customer Name, City, and Age are used to identify the duplicate pair.

Customer 109 is removed after confirming that the details duplicate those of customer 102.

Excel procedure:

1. Select the complete customer dataset.
2. Open the Data tab.
3. Select Remove Duplicates.
4. Choose Customer Name, City, and Age as the comparison columns.
5. Click OK and review the number of removed records.

Customer ID is excluded from the comparison because the duplicate records have different IDs.

Step 3: Handle the missing age

Customer 106 has a missing age. The available ages are:

22, 25, 24, 23, 28, 26, 27, and 30.

Sort the values in ascending order:

22, 23, 24, 25, 26, 27, 28, 30.

There are eight valid age values. The median is the average of the fourth and fifth values:

\[\text{Median}=\frac{25+26}{2}=25.5 \]

For a whole-number age field, this example uses 26 as the replacement value.

Why use the median?

- The median represents the middle of the available age values.
- It is less affected by unusually high or low values than the mean.
- It provides a simple replacement method for this small example.

The replacement should be documented because the original age of the customer is unknown. In a real application, obtaining the correct age from the customer would be preferable.

Step 4: Handle the missing city

Customer 108 has a missing city value.

The original dataset does not provide enough information to determine the customer's actual city. Therefore, the missing city is replaced with “Unknown.”

This approach preserves the customer record without inventing a city.

Step 5: Correct spelling errors

Replace the incorrect city names with standardized values.

- Replace “Banglore” with “Bangalore.”
- Replace “Hydrabad” with “Hyderabad.”

In Excel, the Find and Replace feature can be used:

1. Press Ctrl + H.
2. Enter Banglore in the Find what field.
3. Enter Bangalore in the Replace with field.
4. Select Replace All.
5. Repeat the procedure for Hydrabad and Hyderabad.

Review the resulting dataset to ensure that the corrections have been applied consistently.

Step 6: Validate the cleaned dataset

After cleaning, check the following:

- No duplicate customer details remain under the selected matching rule.
- No required field is left blank in this example.
- City names use standardized spelling.
- Age values are numerical and reasonable for the sample.
- Customer IDs are retained for the remaining records.
- The final number of records is correct.

8. Corrected Customer Dataset

After removing one duplicate record, handling missing values, and correcting spelling errors, the final dataset contains nine records.

Customer ID	Customer Name	City	Age
101	Arun	Chennai	22
102	Priya	Bangalore	25
103	Kumar	Chennai	24
104	Divya	Hyderabad	23
105	Rahul	Mumbai	28
106	Sneha	Chennai	26
107	Vijay	Bangalore	26
108	Meena	Unknown	27
110	Karthik	Mumbai	30

The original number of records was 10. After removing one duplicate record, the final count is 9. Both missing fields have been handled, and the identified spelling errors have been corrected.

9. Before-and-After Comparison

Data Quality Measure	Before Cleaning	After Cleaning
Total records	10	9
Missing age values	1	0
Missing city values	1	0
Duplicate records identified	1 redundant record	0 identified duplicates
Incorrect city spellings	2 incorrect entries	0 identified spelling errors
Unresolved city values	1	1 marked as Unknown

The city “Unknown” is not a spelling error or a confirmed city. It is an explicit indicator that the original information was unavailable. The age 26 is an imputed value, not a verified age.

10. Advantages of Data Cleaning

1. Improves data accuracy: Correcting spelling errors and invalid values improves consistency.
2. Improves data completeness: Handling missing values reduces gaps in important fields.
3. Reduces duplication: Removing redundant records prevents repeated information from distorting reports.
4. Improves analysis: Clean records are easier to group, compare, and summarize.

5. Supports better decisions: Reliable data helps organizations make informed business decisions.
6. Improves reporting: Standardized values prevent the same category from being counted separately under different spellings.
7. Saves time: A clean dataset requires less repeated correction during later analysis.

11. Applications of Data Cleaning

Data cleaning is used in customer relationship management systems, banking databases, hospital information systems, student attendance systems, online shopping platforms, sales databases, and machine-learning projects.

For example, a college may clean its student database by removing duplicate student records, correcting department names, and handling missing contact information. This improves the reliability of attendance reports and student statistics.

Similarly, an online shopping company may clean its customer and transaction data before analyzing customer preferences or developing recommendation systems.

12. Limitations and Precautions

- Replacing missing values can introduce uncertainty because the original information is not known.
- Removing duplicates without checking customer identity can accidentally delete valid records.
- Replacing missing categorical values with “Unknown” preserves the record but does not recover the missing information.
- Automatic spelling correction may produce errors when two different names are similar.
- All cleaning decisions should be documented so that the dataset can be reviewed and the changes explained.

Result

The customer dataset was cleaned successfully by identifying missing values, detecting and removing a duplicate customer record, and correcting spelling errors. The missing age was replaced with the median-based value of 26, and the missing city was marked as “Unknown.” The final dataset contains nine records and is more consistent and suitable for further analysis.